

delta_and_hatdelta_periods_executed

May 13, 2026

1 Periods and Quasi-Periods of Δ and $\hat{\Delta}$

Reproduce Brown's numerical values (1710.07912 §8.3) using the **finite-endpoint Diamantis-Rolen cocycle formula**:

$$\sigma_f(\gamma)(z) = \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau)(\tau - z)^{k-2} d\tau$$

Targets: - Δ : $\omega^+ = -68916772.809595194754 \dots$, $\omega^- = -5585015.3793104018668 \dots$ - $\hat{\Delta} = \Delta(J^2 + 24J - 393444)$: $\eta^+ = 127202100647.17709477 \dots$, $\eta^- = 10276732343.649132750 \dots$

Procedure: 1. Fix $\tau_0 \in \mathbb{H}$ (we use $\tau_0 = 0.3 + 1.2i$). 2. Compute $C_\gamma(X, Y) = (2\pi i)^{11} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau)(X - \tau Y)^{10} d\tau$ for $\gamma \in \{S, T\}$. 3. Solve $P|_T - P = C_T$ for the coboundary polynomial $P \in V_{10}$. 4. Form $C'_S = C_S - (P|_S - P)$. 5. Decompose $C'_S = \omega^+ P_S^+ + \omega^- P_S^- \pmod{X^{10} - Y^{10}}$ junk direction). 6. Read off periods.

All integration is over **finite-length line segments in $\mathbb{H}$** . Nothing approaches the cusp.

```
[1]: from mpmath import mp, mpc, mpf, pi, exp, quad, matrix, lu_solve
from sympy import Rational
from math import comb
import time

mp.dps = 40
n = 10
TWOPII = mpc(0, 2*pi)
PREFACTOR = TWOPII**(n+1)
print(f'Working at {mp.dps} decimal digits.')
```

Working at 40 decimal digits.

1.1 1. Build q -series for Δ , j , $J = j - 744$, and $\hat{\Delta}$

$\hat{\Delta} = \Delta \cdot (J^2 + a_1 J + a_0)$, with a_1, a_0 determined by demanding $\hat{\Delta} = q^{-1} + 0 + 0 \cdot q + O(q^2)$.

Since Δ has $a_0 = 0$ and $a_1 = 1$, the linear conditions read: - q^0 of $\hat{\Delta} = 0$: $[\Delta J^2]_0 + a_1 [\Delta J]_0 = 0$. - q^1 of $\hat{\Delta} = 0$: $[\Delta J^2]_1 + a_1 [\Delta J]_1 + a_0 = 0$.

We work with exact rationals.

```

[2]: PREC_QSERIES = 60

def laurent_mul(a, b, lo, hi):
    out = {}
    for ka, va in a.items():
        for kb, vb in b.items():
            nn = ka + kb
            if lo <= nn <= hi:
                out[nn] = out.get(nn, Rational(0)) + va * vb
    return {k: v for k, v in out.items() if v != 0}

def laurent_div(num, den, lo_exp, hi_exp):
    den_min = min(den.keys())
    den_lead = den[den_min]
    quot = {}
    for target_n in range(lo_exp + den_min, hi_exp + den_min + 1):
        s = num.get(target_n, Rational(0))
        for k_quot in range(lo_exp, target_n - den_min):
            s -= quot.get(k_quot, Rational(0)) * den.get(target_n - k_quot,
↪Rational(0))
        if s != 0:
            quot[target_n - den_min] = s / den_lead
    return quot

def divisor_sigma(nn, k):
    s = 0; d = 1
    while d*d <= nn:
        if nn % d == 0:
            s += d**k
            if d != nn // d:
                s += (nn // d)**k
        d += 1
    return s

# Delta = q * prod (1-q^n)^{24}
def eta24_qexp(prec):
    series = {0: Rational(1)}
    for nn in range(1, prec + 1):
        factor = {}
        for k in range(25):
            te = nn * k
            if te > prec: break
            factor[te] = Rational((-1)**k * comb(24, k))
        series = laurent_mul(series, factor, 0, prec)
    return series

eta24 = eta24_qexp(PREC_QSERIES + 2)

```

```

Delta_series = {k+1: v for k, v in eta24.items() if k+1 <= PREC_QSERIES + 2}

# E4, j, J
E4 = {0: Rational(1)}
for nn in range(1, PREC_QSERIES + 3):
    E4[nn] = Rational(240 * divisor_sigma(nn, 3))
E4_cubed = laurent_mul(laurent_mul(E4, E4, 0, PREC_QSERIES + 2), E4, 0,
    ↪PREC_QSERIES + 2)
j_series = laurent_div(E4_cubed, Delta_series, -1, PREC_QSERIES)
J_series = dict(j_series); J_series[0] = J_series.get(0, Rational(0)) - 744
J_series = {k: v for k, v in J_series.items() if v != 0}
J_sq = laurent_mul(J_series, J_series, -2, PREC_QSERIES)

print('Delta:', {k: Delta_series[k] for k in sorted(Delta_series)[:5]})
print('J = j - 744:', {k: J_series[k] for k in sorted(J_series)[:5]})
print('J^2:', {k: J_sq[k] for k in sorted(J_sq)[:5]})

```

```

Delta: {1: 1, 2: -24, 3: 252, 4: -1472, 5: 4830}
J = j - 744: {-1: 1, 1: 196884, 2: 21493760, 3: 864299970, 4: 20245856256}
J^2: {-2: 1, 0: 393768, 1: 42987520, 2: 40491909396, 3: 8504046600192}

```

```

[3]: # Solve for a_1, a_0 in hat_Delta = Delta * (J^2 + a_1 J + a_0)
D_Jsq = laurent_mul(Delta_series, J_sq, -2, PREC_QSERIES)
D_J    = laurent_mul(Delta_series, J_series, -1, PREC_QSERIES)

# q^0 eqn: D_Jsq[0] + a_1 * D_J[0] = 0
a_1 = -D_Jsq[0] / D_J[0]
# q^1 eqn: D_Jsq[1] + a_1 D_J[1] + a_0 * Delta[1] = 0 (Delta[1] = 1)
a_0 = -(D_Jsq[1] + a_1 * D_J[1])
print(f'a_1 = {a_1}, a_0 = {a_0}')
assert a_1 == 24 and a_0 == -393444

# hat_Delta
hat_Delta_series = {}
for k, v in D_Jsq.items():
    hat_Delta_series[k] = hat_Delta_series.get(k, Rational(0)) + v
for k, v in D_J.items():
    hat_Delta_series[k] = hat_Delta_series.get(k, Rational(0)) + a_1 * v
for k, v in Delta_series.items():
    hat_Delta_series[k] = hat_Delta_series.get(k, Rational(0)) + a_0 * v
hat_Delta_series = {k: v for k, v in hat_Delta_series.items() if v != 0}

print('\nhat_Delta first coefficients:')
for k in sorted(hat_Delta_series)[:8]:
    print(f' q^{k}: {hat_Delta_series[k]}')
print('\nBrown reports: q^{-1}=1, q^2=47709536, q^3=39862705122,
    ↪q^4=7552626810624')

```

```
a_1 = 24, a_0 = -393444
```

```
hat_Delta first coefficients:
```

```
q^-1: 1
q^2: 47709536
q^3: 39862705122
q^4: 7552626810624
q^5: 609136463852480
q^6: 27947672851540608
q^7: 855425832767342991
q^8: 19261421726562785280
```

```
Brown reports: q^{-1}=1, q^2=47709536, q^3=39862705122, q^4=7552626810624
```

1.2 2. Numeric evaluators for Δ and $\hat{\Delta}$

Δ via the η^{24} product (fast convergence). $\hat{\Delta}$ via its truncated q -expansion (also fast since $|q| \sim 10^{-3}$ on our paths).

```
[4]: def delta_form(tau, N_eta=300):
    tau = mpc(tau)
    q = exp(TWOPII * tau)
    log_prod = mpc(0)
    qn = q
    for _ in range(N_eta):
        log_prod += mp.log(1 - qn)
        qn *= q
    eta = exp(TWOPII*tau/24 + log_prod)
    return eta**24

# Convert hat_Delta to mpf coefficients
hat_coeffs_mp = sorted([(int(e), mpf(int(c))) for e, c in hat_Delta_series.
    ↪items()])

def hat_delta(tau):
    tau = mpc(tau)
    q = exp(TWOPII * tau)
    val = mpc(0)
    for e, c in hat_coeffs_mp:
        val += c * (q ** e)
    return val

# Quick checks
tau_test = mpc(0, 1)
print(f'Delta(i) = {delta_form(tau_test)}')
print(f'hat_Delta(i) = {hat_delta(tau_test)}')
```

```
Delta(i) = (0.001785369850642151904343054960342262310581 + 0.0j)
hat_Delta(i) = (1068.415308980081679812205693807699349996 + 0.0j)
```

1.3 3. Cocycle integration and slash operators

Generic over the form f : pass in any callable $f(\tau)$ returning the numeric value. We evaluate

$$\int_{\tau_a}^{\tau_b} f(\tau)(X - \tau Y)^{10} d\tau$$

as a polynomial in X, Y along a straight-line path in $\mathbb{H}$.

```
[5]: def cocycle_polynomial(f, tau_a, tau_b, n_max=10):
    """(2 pi i)^{n+1} int_{tau_a}^{tau_b} f(tau) (X - tau Y)^n dtau, as a
    ↪coefficient list for X^{n-j}Y^j."""
    tau_a = mpc(tau_a); tau_b = mpc(tau_b)
    dtau = tau_b - tau_a
    moments = []
    for n_pow in range(n_max + 1):
        def integrand(s, n_pow=n_pow):
            tau = tau_a + s * dtau
            return f(tau) * tau**n_pow * dtau
        moments.append(quad(integrand, [0, 1]))
    coeffs = []
    for j in range(n_max + 1):
        c = mpf(comb(n_max, j)) * mpf((-1)**j) * moments[j]
        coeffs.append(PREFACTOR * c)
    return coeffs

def slash_T(p):
    n_loc = len(p) - 1
    out = [mpf(0)] * (n_loc + 1)
    for j in range(n_loc + 1):
        for i in range(n_loc - j + 1):
            out[j + i] += p[j] * mpf(comb(n_loc - j, i))
    return out

def slash_S(p):
    n_loc = len(p) - 1
    return [p[n_loc - m] * mpf((-1)**m) for m in range(n_loc + 1)]
```

1.4 4. Generic period extraction routine

Wraps the whole pipeline: feed in f and τ_0 , get (ω^+, ω^-) (or (η^+, η^-)).

```
[6]: # Brown's basis (1710.07912 §8.2)
P_plus = [mpf(0)]*11
P_plus[0] = -mpf(36)/mpf(691); P_plus[10] = mpf(36)/mpf(691)
```

```

P_plus[2] = mpf(1); P_plus[4] = -mpf(3); P_plus[6] = mpf(3); P_plus[8] = -
↳-mpf(1)
P_minus = [mpf(0)]*11
P_minus[1] = mpf(4); P_minus[3] = -mpf(25); P_minus[5] = mpf(42)
P_minus[7] = -mpf(25); P_minus[9] = mpf(4)

def extract_periods(f, tau0, label='f', verbose=True):
    if verbose:
        print(f'\n===== Extracting periods of {label} at tau_0 = {tau0}↳
↳=====')
    t0 = time.time()
    C_S = cocycle_polynomial(f, -1/tau0, tau0, n)
    if verbose: print(f' C_S done ({time.time()-t0:.1f} s)')
    t0 = time.time()
    C_T = cocycle_polynomial(f, tau0 - 1, tau0, n)
    if verbose: print(f' C_T done ({time.time()-t0:.1f} s)')

    if verbose:
        print(f' |C_T|_max = {max(abs(c) for c in C_T)}')
        print(f' [C_T]_X^10 = {abs(C_T[0])} (cuspidality, should be ~0)')

    # Solve P/_T - P = C_T for P
    A = matrix(10, 10); b = matrix(10, 1)
    for m in range(1, 11):
        for j in range(0, m):
            A[m-1, j] = mpf(comb(10 - j, m - j))
        b[m-1, 0] = C_T[m]
    psol = lu_solve(A, b)
    p_coeffs = [psol[j, 0] for j in range(10)] + [mpf(0)]

    PS = slash_S(p_coeffs)
    C_S_prime = [C_S[j] - (PS[j] - p_coeffs[j]) for j in range(11)]

    # Internal consistency: read off from each available monomial
    minus_estimates = [C_S_prime[j] / P_minus[j] for j in [1,3,5,7,9]]
    plus_estimates = [C_S_prime[j] / P_plus[j] for j in [2,4,6,8]]

    if verbose:
        print(' - estimates (odd):')
        for j, e in zip([1,3,5,7,9], minus_estimates):
            print(f' X^{10-j}Y^{j}: {e}')
        print(' + estimates (even, interior):')
        for j, e in zip([2,4,6,8], plus_estimates):
            print(f' X^{10-j}Y^{j}: {e}')

    return plus_estimates[0], minus_estimates[0]

```

1.5 5. Run for Δ

```
[7]: tau0 = mpc('0.3', '1.2')
omega_plus, omega_minus = extract_periods(delta_form, tau0, 'Delta')
print('\n=== EXTRACTED ===')
print('omega^+ =', omega_plus)
print('omega^- =', omega_minus)
print('\n=== BROWN ===')
print('omega^+ = -68916772.809595194754...')
print('omega^- = -5585015.3793104018668...')
print('\n=== RATIOS (i factor expected on omega^-) ===')
print('omega^+ / Brown:', omega_plus / mpf('-68916772.809595194754'))
print('omega^- / Brown:', omega_minus / mpf('-5585015.3793104018668'))
```

```
===== Extracting periods of Delta at tau_0 = (0.3 + 1.2j) =====
C_S done (17.5 s)
C_T done (17.5 s)
|C_T|_max = 234053334.6157587586133717302731708676396
[C_T]_X^10 = 4.492405684810882055780909631912922994899e-38 (cuspidality,
should be ~0)
```

```
- estimates (odd):
X^9Y^1: (-2.407412430484044816319972428231159148173e-35 -
5585015.379310401866877139263796275129635j)
X^7Y^3: (6.162975822039154729779129416271767419322e-35 -
5585015.379310401866877139263796275129635j)
X^5Y^5: (3.668437989309020672487577033495099654358e-35 -
5585015.379310401866877139263796275129635j)
X^3Y^7: (9.244463733058732094668694124407651128983e-35 -
5585015.379310401866877139263796275129635j)
X^1Y^9: (0.0 - 5585015.379310401866877139263796275129635j)
+ estimates (even, interior):
X^8Y^2: (-68916772.80959519475431012465533103043907 -
3.851859888774471706111955885169854637076e-34j)
X^6Y^4: (-68916772.80959519475431012465533103043907 +
5.135813185032628941482607846893139516102e-34j)
X^4Y^6: (-68916772.80959519475431012465533103043907 + 0.0j)
X^2Y^8: (-68916772.80959519475431012465533103043907 -
7.703719777548943412223911770339709274152e-34j)
```

```
=== EXTRACTED ===
omega^+ = (-68916772.80959519475431012465533103043907 -
3.851859888774471706111955885169854637076e-34j)
omega^- = (-2.407412430484044816319972428231159148173e-35 -
5585015.379310401866877139263796275129635j)
```

```
=== BROWN ===
omega^+ = -68916772.809595194754...
```

```
=== RATIOS (i factor expected on omega^-) ===  
omega^+ / Brown: (1.00000000000000000004499988068040414353 +  
5.589147215898336646516247831747089732221e-42j)  
omega^- / Brown: (4.310484872436098910871485948806665842545e-42 +  
1.00000000000000000001381182656757513521j)
```

Same procedure, same τ_0 , just a different f .

```

===== Extracting periods of hat_Delta at tau_0 = (0.3 + 1.2j) =====
C_S done (16.3 s)
C_T done (19.1 s)
|C_T|_max = 200673146409032.1503042254499963791663889
[C_T]_X^10 = 1.55711272805288252882114348751555546029e-31 (cuspidality, should
be ~0)

- estimates (odd):
X^9Y^1: (8.83524213847533222061135993116720577234e-29 +
10276732343.64913275081719307240092090892j)
X^7Y^3: (9.693522803355793064899320610194877190224e-29 +
10276732343.64913275081719307240092090896j)
X^5Y^5: (-4.327465537212407618258625272408427317064e-29 +
10276732343.64913275081719307240092090903j)
X^3Y^7: (0.0 + 10276732343.64913275081719307240092090893j)
X^1Y^9: (-2.524354896707237777317531408904915934954e-29 +
10276732343.6491327508171930724009209089j)

+ estimates (even, interior):
X^8Y^2: (127202100647.1770947773171612986108774946 -
1.009741958682895110927012563561966373982e-27j)
X^6Y^4: (127202100647.1770947773171612986108774957 -
6.731613057885967406180083757079775826545e-28j)
X^4Y^6: (127202100647.1770947773171612986108774951 + 0.0j)
X^2Y^8: (127202100647.1770947773171612986108774938 + 0.0j)

```

=== EXTRACTED ===

```
eta^+ = (127202100647.1770947773171612986108774946 -
1.009741958682895110927012563561966373982e-27j)
eta^- = (8.83524213847533222061135993116720577234e-29 +
10276732343.64913275081719307240092090892j)
```

=== BROWN ===

```
eta^+ = 127202100647.17709477...
eta^- = 10276732343.649132750...
```

=== RATIOS (i factor expected on eta^-) ===

```
eta^+ / Brown: (1.00000000000000000057523902996748678468 -
7.938091851829049113519975280144469657463e-39j)
eta^- / Brown: (8.59732631251740251091035137966596707041e-39 +
1.00000000000000000079518765797761973862j)
```

1.7 7. Basepoint independence sanity check

Cohomology class shouldn't depend on τ_0 . Pick a different one and verify.

```
[9]: tau0_alt = mpc('-0.2', '0.9')
eta_plus_alt, eta_minus_alt = extract_periods(hat_delta, tau0_alt, 'hat_Delta_
↳(alt tau_0)', verbose=False)
print(f'eta^+ at tau_0 = {tau0}: {eta_plus}')
print(f'eta^+ at tau_0 = {tau0_alt}: {eta_plus_alt}')
print(f' difference: {eta_plus - eta_plus_alt}')
print()
print(f'eta^- at tau_0 = {tau0}: {eta_minus}')
print(f'eta^- at tau_0 = {tau0_alt}: {eta_minus_alt}')
print(f' difference: {eta_minus - eta_minus_alt}')
```

```
eta^+ at tau_0 = (0.3 + 1.2j): (127202100647.1770947773171612986108774946 -
1.009741958682895110927012563561966373982e-27j)
eta^+ at tau_0 = (-0.2 + 0.9j): (127202100647.1770947773171612986108774935 -
6.05845175209737066556207538137179824389e-28j)
difference: (1.11071615455118462201971381991816301138e-27 -
4.038967834731580443708050254247865495927e-28j)
```

```
eta^- at tau_0 = (0.3 + 1.2j): (8.83524213847533222061135993116720577234e-29 +
10276732343.64913275081719307240092090892j)
eta^- at tau_0 = (-0.2 + 0.9j): (-3.786532345060856665976297113357373902431e-29
+ 10276732343.64913275081719307240092090888j)
difference: (1.262177448353618888658765704452457967477e-28 +
4.41762106923766611030567996558360288617e-29j)
```

1.8 Summary of conventions used

- **Cocycle:** $C_\gamma(X, Y) = (2\pi i)^{11} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau)(X - \tau Y)^{10} d\tau$. Direct port of Diamantis-Rolen eq. (2.16) bottom formula, with cusp form normalization.

- **Slash actions:** $P|_T(X, Y) = P(X + Y, Y)$, $P|_S(X, Y) = P(Y, -X)$.
- **Coboundary modification:** solve $P|_T - P = C_T$ for P , then $C'_S = C_S - (P|_S - P)$.
- **Brown basis** (Brown 1710.07912 §8.2): $P_S^+ = \frac{36}{691}(Y^{10} - X^{10}) + X^2Y^2(X^2 - Y^2)^3$, $P_S^- = 4X^9Y - 25X^7Y^3 + 42X^5Y^5 - 25X^3Y^7 + 4XY^9$.

Convention discrepancy: extracted ω^-, η^- come out as i times Brown's reported real values. This matches Brown's own period-matrix convention in eq. (2.13) where the entries are $\omega^+, i\omega^-$ — i.e., he absorbs the i into the basis on one side and not the other depending on context. The magnitudes match to the full reported precision.